



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
10-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 192

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

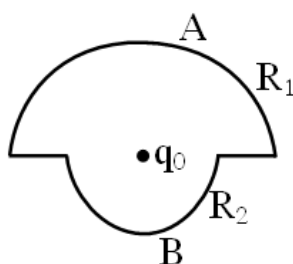
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A charge particle q_0 is kept at centre of 2 non conducting hemi spherical shell of radius R_1 & R_2 . If E_1 & E_2 is electric field at surfaces of A & B then



(A) Ratio of flux $\frac{\phi_1}{\phi_2} = 1$.

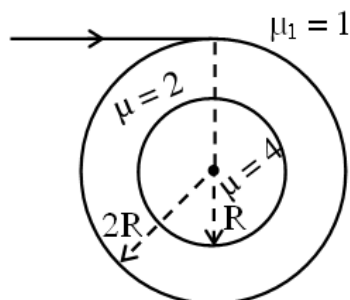
(B) $E_1 = E_2$

(C) $\frac{R_1}{R_2} = \sqrt{\frac{E_2}{E_1}}$

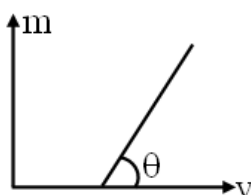
(D) $E_1 \neq E_2$

Space for Rough Work

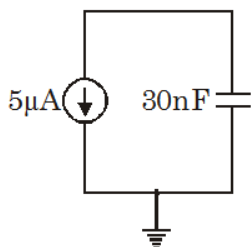
2. A ray of light tangentially incident on outer sphere.



- (A) Net deviation suffered by ray is 240° .
 (B) Ray will never enter inside inner sphere.
 (C) Ray is incident tangentially on inner sphere.
 (D) Angle made by ray with normal inside inner sphere is 30° .
3. Variation of magnification of convex lens vs image distance for real objects is shown. Choose correct statement.



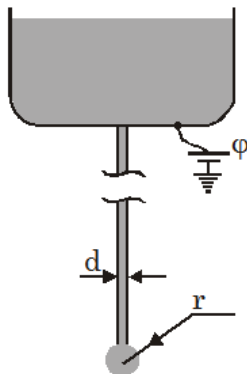
- (A) f is equal to intercept of V -axis. (B) f is equal to $\frac{1}{\text{slope}}$ of graph.
 (C) f is equal to slope of graph (D) magnitude of intercept on m axis is 1.
4. Diagram shows a capacitor connected with a constant current source. Choose the correct option(s) :



- (A) the rate of change of voltage across the capacitor in the circuit of figure from top to bottom 166.7 V/s.
 (B) the rate of change of voltage across the capacitor in the circuit of figure from top to bottom -166.7 V/s.
 (C) the capacitor's voltage 10 milliseconds after power is switched on is 1.667V.
 (D) the capacitor's voltage 10 milliseconds after power is switched on is 2.667V.

Space for Rough Work

5. A long metallic pipe with internal diameter d is pointing directly downwards. Water is slowly dripping from a nozzle at its lower end, see fig. Water can be considered to be electrically conducting; its surface tension is σ and its density is ρ . A droplet of radius r hangs below the nozzle. The radius grows slowly in time until the droplet separates from the nozzle due to the free fall acceleration g . Always assume that $d \ll r$. Choose the correct option(s) :-

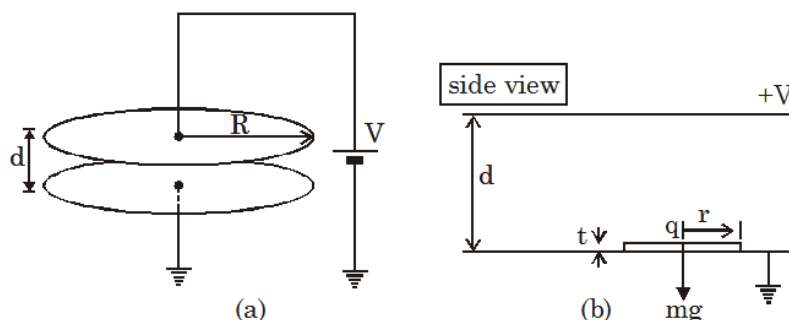


- (A) the radius $r_{\max}$ of a drop just before it separates from the nozzle $\sqrt[3]{\frac{1}{4} \frac{\sigma d}{\rho g}}$.
- (B) the radius $r_{\max}$ of a drop just before it separates from the nozzle $\sqrt[3]{\frac{3}{4} \frac{\sigma d}{\rho g}}$.
- (C) Relative to the far-away surroundings, the pipe's electrostatic potential is ϕ . the charge Q of a drop when its radius is r , $Q = 4\pi\epsilon_0\phi r$
- (D) Consider the situation in which r is kept constant and ϕ is slowly increased. The droplet becomes unstable and breaks into pieces if the hydrostatic pressure inside the droplet becomes smaller than the atmospheric pressure, the critical potential $\phi_{\max}$ at which this will happen $2\sqrt{\frac{\sigma r}{\epsilon_0}}$.

Space for Rough Work

6. A capacitor consists of two circular parallel plates both with radius R separated by distance d , where $d \ll R$, as shown in figure (a). The top plate is connected to a constant voltage source at a potential V while the bottom plate is grounded. Then a thin and small disk of mass m with radius r ($\ll R, d$) and thickness t ($\ll r$) is placed on the center of the bottom plate, as shown in figure (b).

Let us assume that the space between the plates is in vacuum with the dielectric constant ϵ_0 ; the plates and the disk are made of perfect conductors; and all the electrostatic edge effects may be neglected. The inductance of the whole circuit and the relativistic effects can be safely disregarded. The image charge effect can also be neglected. Choose the correct option(s)



- (A) When the disk is placed on the bottom plate, a charge q on the disk of figure (b) is related to the voltage V by $q = \chi V$, then χ in terms of r, d and ϵ_0 is given by $-\frac{\epsilon_0 \pi r^2}{d}$.
- (B) When the disk is placed on the bottom plate, a charge q on the disk of figure (b) is related to the voltage V by $q = \chi V$, then χ in terms of r, d and ϵ_0 is given by $\frac{\epsilon_0 \pi r^2}{d}$.
- (C) The parallel plates lie perpendicular to a uniform gravitational field g . To lift up the disk at rest initially, we need to increase the applied voltage beyond a threshold voltage v_{th} . The value of v_{th} is given by $\sqrt{\frac{2mgd}{\chi}}$.
- (D) The parallel plates lie perpendicular to a uniform gravitational field g . To lift up the disk at rest initially, we need to increase the applied voltage beyond a threshold voltage v_{th} . The value of v_{th} is given by $\sqrt{\frac{mgd}{\chi}}$.

Space for Rough Work

7. A massive star of mass M is in uniform circular orbit around a supermassive black hole of mass M_b . Initially, the radius and angular frequency of the orbit are R and ω respectively. According to Einstein's theory of general relativity the space around the two objects is distorted and gravitational waves are radiated. Energy is lost through this radiation and as a result the orbit of the star shrinks gradually. One may assume, however, that the orbit remains circular throughout and Newtonian mechanics holds. Choose the correct option(s) :-

(A) The power radiated through gravitational wave by this star is given by $L_G = KC^x G^y M^2 R^4 \omega^6$ where c is the speed of light, G is the universal gravitational constant, and K is a dimensionless constant, then $x = -5$ & $y = 1$

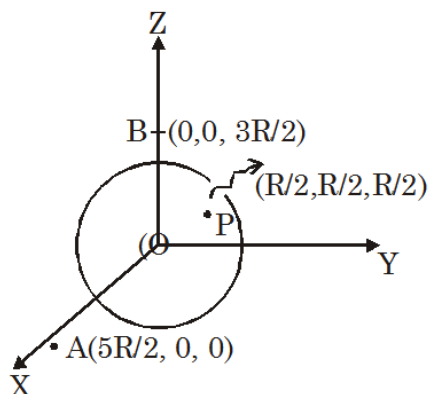
(B) Total mechanical energy of the star-black hole system is $\frac{GMm_b}{2R}$

(C) An expression for the rate of decrease in the orbital period (dT/dt) in terms of the masses, period T and constants given by $\frac{dT}{dt} = -\frac{3KG^{5/3}M_b^{2/3}M(2\pi)^{8/3}}{c^5 T^{5/3}}$

(D) An expression for the rate of decrease in the orbital period (dT/dt) in terms of the masses, period T and constants given by $\frac{dT}{dt} = \frac{3KG^{5/3}M_b^{2/3}M}{c^5 T^{5/3}}$

Space for Rough Work

8. An uncharged solid spherical conductor of radius R has its centre at the origin of a cartesian frame OXY Z. Two point charges each of $+Q$ are placed at fixed positions at $(5R/2, 0, 0)$ and $(0, 0, 3R/2)$. There will be induced charges on the sphere due to these charges. Choose the correct option(s)



- (A) Electric field at $\left(\frac{R}{2}, \frac{R}{2}, \frac{R}{2}\right)$ due to induced charges is given by

$$E_P = -\frac{2^{3/2}Q}{(4\pi\epsilon_0 R^2)} \left(\frac{9}{2} - 2\sqrt{3}\right) \hat{i} + \left(\frac{9}{2} + \sqrt{3}\right) \hat{j} + (9 - \sqrt{3}) \hat{k}$$

- (B) Electric field at $\left(\frac{R}{2}, \frac{R}{2}, \frac{R}{2}\right)$ due to induced charges is given by

$$E_P = -\frac{2^{3/2}Q}{(4\pi\epsilon_0 R^2)} \left(\frac{9}{2} - 2\sqrt{3}\right) \hat{i} + \left(\frac{9}{2} + \sqrt{3}\right) \hat{j} + (9 + \sqrt{3}) \hat{k}$$

- (C) The potential at P due to induced charges is given by

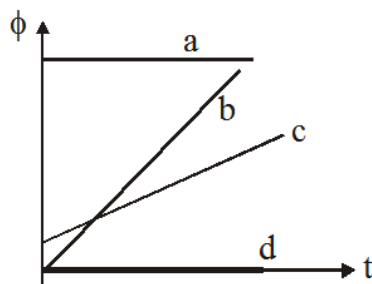
$$V = \frac{Q}{(4\pi\epsilon_0 R) \left(\frac{16}{15} - \left(\frac{2}{9}\right)^{1/2} - \left(\frac{2}{3}\right)^{1/2}\right)}$$

- (D) The potential at P due to induced charges is given by

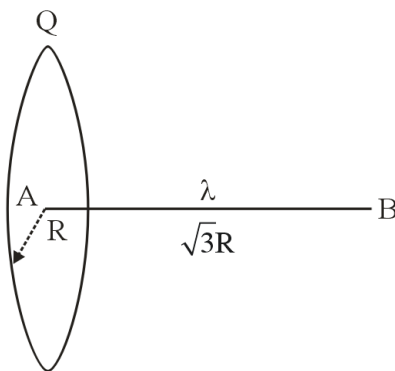
$$V = \frac{Q}{(4\pi\epsilon_0 R) \left(\frac{16}{15} + \left(\frac{2}{9}\right)^{1/2} + \left(\frac{2}{3}\right)^{1/2}\right)}$$

Space for Rough Work

9. The graph shows variation of flux (ϕ) with time (t) passing through a sphere for various conditions. Choose the correct option(s) : (Assume : linear charge on x-axis)



- (A) Graph 'a' may correspond to the case when a sphere is moving along x-axis with constant speed and the charge distribution is uniform in x-direction.
- (B) Graph 'b' may correspond to the case when a sphere is moving along x-axis with constant speed and the charge distribution is proportional to x .
- (C) Graph 'c' may correspond to the case when a sphere is moving along x-axis with constant speed and the charge distribution is varying linearly with x .
- (D) Graph 'd' may correspond to the case when a sphere is located at origin and its radius is increasing with uniform rate and charge distribution is proportional to x .
10. A non-conducting rod AB of length $\sqrt{3}R$, uniformly distributed charge of linear charge density λ and a non-conducting ring of uniformly distributed charge Q , are placed as shown in the figure. Point A is the centre of ring and line AB is the axis of the ring, perpendicular to plane of ring. The electrostatic interaction energy between ring and rod is:



- (A) $\frac{Q\lambda}{4\pi\epsilon_0} \ln(2 + \sqrt{3})$ (B) $\frac{Q\lambda}{2\pi\epsilon_0} \ln(2 + \sqrt{3})$ (C) $\frac{Q\lambda}{4\pi\epsilon_0} \ln(2 - \sqrt{3})$ (D) $\frac{Q\lambda}{2\pi\epsilon_0} \ln(2 - \sqrt{3})$

Space for Rough Work

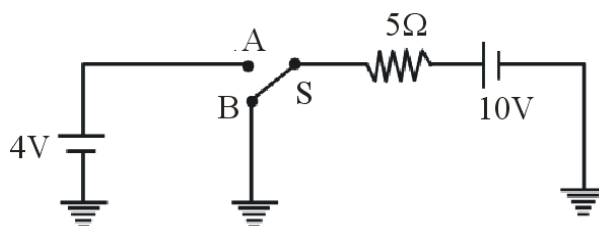
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

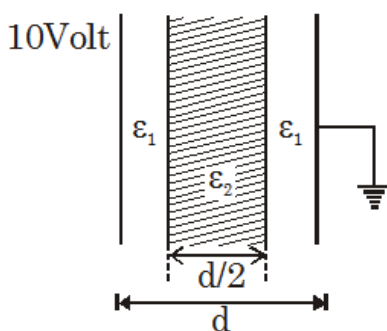
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. In given circuit when switch S is in position A current in resistance is i_1 & position B it is i_2 , then $\frac{i_1}{i_2}$ is equal to:



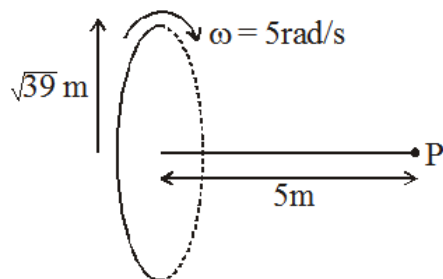
2. x-z plane separates 2 medium A & B having $\mu_1 = 1.5$ & $\mu_2 = 2$. A ray of light passes through two medium from A to B. If unit vector of ray are $a\hat{i} + b\hat{j}$ & $c\hat{i} + d\hat{j}$ in A & B. then value of a/c is
3. A parallel plate capacitor consisting two dielectric materials is shown in the figure. The middle dielectric slab is place symmetrically with respect to the plates. If the potential difference between one of the plates and the nearest surface of dielectric interface is 2Volts, then the ratio $\epsilon_1 : \epsilon_2$ is



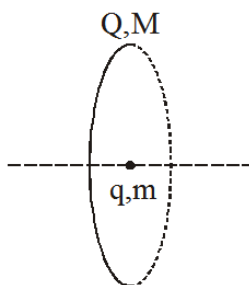
4. Two ammeters X and Y have resistances of 1.2Ω and 1.5Ω respectively and they give full scale deflection with 150 mA and 250 mA respectively. The ranges have been extended by connecting shunts so as to give full scale deflection with 15 A. The ammeters along with shunts are connected in parallel and then placed in a circuit in which the total current flowing is 15A. The current in amperes indicated in ammeter X is.

Space for Rough Work

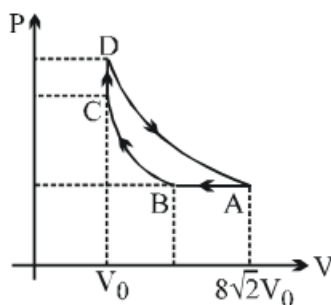
5. Figure shows a uniformly charged disc of radius $\sqrt{39}$ m and charge density $3 \mu\text{C}/\text{m}^2$ rotating about its axis with angular velocity 5 rad/sec . Find out the magnetic field at a distance 5 m from the center of disc on its axis. Express your answer in pT.



6. Figure shows a ring of mass 'M' and radius 'R' uniformly charged with charge Q. A point charge '-q' of mass m is kept at the centre of ring. Ring is free to move. Find the time period of small oscillation of the point charge. [Take : $m = 1 \text{ kg}$, $M = 2 \text{ kg}$, $q = -3 \mu\text{C}$, $Q = 40 \mu\text{C}$, $R = 2 \text{ m}$]



7. A analog voltmeter uses external multiplier settings (an external resistance connected in series with voltmeter). With a multiplier setting of $20 \text{ k}\Omega$, it reads 440 V and with a multiplier setting of $80 \text{ k}\Omega$ it reads 352 V . For a multiplier setting of $40 \text{ k}\Omega$, find the reading (in V) of voltmeter
8. Helium is used as working substance in an engine working on the cycle as shown in the figure. Processes AB, BC, CD and DA are isobaric, adiabatic, isochoric and isothermal respectively. The ratio of maximum to minimum volume of helium during the cycle is $8\sqrt{2}$ and that of maximum to minimum absolute temperature is 4. Calculate efficiency (η) of the cycle ($\eta < 1$).



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following statement(s) is/are **CORRECT**?

- (A) In the bessemerization process of copper extraction from copper pyrite, ferrous silicate is formed in the form of slag
- (B) In hydrometallurgical process used for the extraction of copper from low grade ore, Fe is commercially used as reducing agent in displacement reaction but not Zinc
- (C) Se and Te is present in the anode mud during electrorefining of impure copper
- (D) Partial roasting of chalcopryrite produces $\text{Cu}_2\text{O} + \text{FeS}$

2. Select the incorrect statements (s) -

- (A) Combustion of methane in an adiabatic rigid container will cause no change in temperature of system.
- (B) It is possible to have both adiabatic reversible and adiabatic irreversible process between two fixed states.
- (C) For reaction involving only ideal gas, occurring at constant temperature there will be no change in internal energy.
- (D) No process is possible for the gases in which volume is changed but PV work is zero.

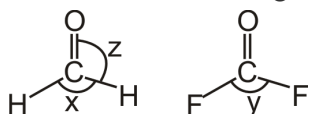
Space for Rough Work

3. In an evacuated vessel of 8.2 L capacity, 0.2 moles of A(g) is taken, vessel is sealed and heated to 127°C, by which the following first order reaction occurred at constant volume and temperature :-

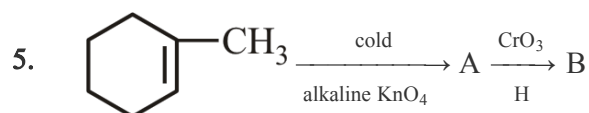


The vapour pressure of C(s) is 0.1 atm at 127°C. If the half life of reaction is 20 min, then the correct information(s) related with reaction is/are : (R = 0.082 L-atm/K-mol)

- (A) $P_{\text{total}} = 1.7 \text{ atm}$ at $t = \infty$
 (B) $P_{\text{total}} = 0.98 \text{ atm}$ after 10% decomposition of A(g)
 (C) $P_{\text{total}} = 0.88 \text{ atm}$ after 5% decomposition of A(g)
 (D) $P_{\text{total}} = 1.3 \text{ atm}$ at $t = 20 \text{ min}$
4. Which of the following relationship is(are) correct for the following figure.

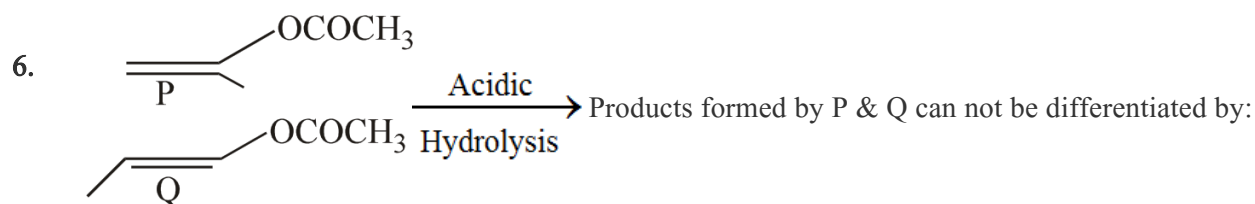


- (A) $x = y = z$ (B) $x = y > z$ (C) $z > 120^\circ$ (D) $x > y$



A and B are :-

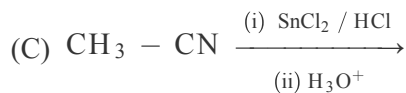
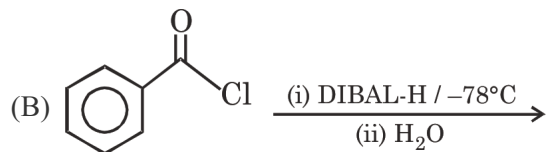
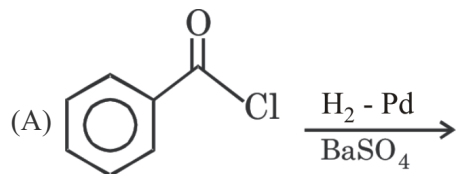
- (A)
- (B)
- (C)
- (D) No formation of A and B

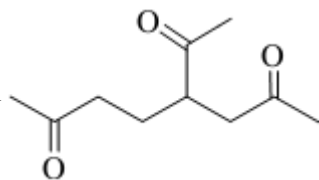


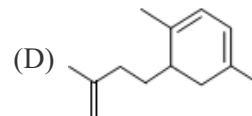
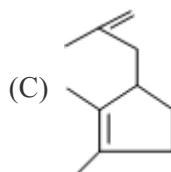
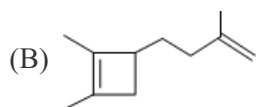
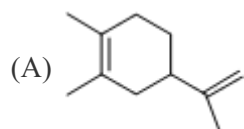
- (A) 2, 4 DNP (B) Lucas reagent (ZnCl_2) conc. HCl
 (C) NaHSO_3 (D) Fehlings solution

Space for Rough Work

7. Which of the following reaction(s) produce aldehyde as a product :



8. Which of the following compound can only form  + HCHO on ozonolysis :



Space for Rough Work

9. Which of the following statement(s) is/are **INCORRECT**?
- (A) H_2S when passed in sodium nitroprusside solution then purple complex is formed
- (B) H_2S reacts with H_2SO_3 solution and produce sulphur, in this reaction H_2S acts as oxidizing agent
- (C) NaNO_2 produce a basic gas on reaction with Zn dust + KOH solution
- (D) NaNO_2 give red coloured product when treated with sulphanilic acid and dil. CH_3COOH followed by the addition of 1-naphthylamine.
10. Light from a discharge tube containing hydrogen atom fall on the surface of a piece of sodium. The kinetic energy of a fastest photo electron emitted from sodium is 0.73 eV and work function for sodium is 1.82 eV. Collector plate is connected to +ve terminal of battery having voltage 2 volts. Which of the following are true regarding above experiment :
- (A) The energy of photon causing the photoelectric effect is 2.55 eV.
- (B) Hydrogen atom electron undergoes a transition from $n = 4$ to $n = 2$ to show given photoelectric effect.
- (C) Change in angular momentum of the electron in the hydrogen atom will be h/π .
- (D) Kinetic energy of photoelectron at emitter plate will be 0.73 eV and at collector plate will be 2.73 eV.
-

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

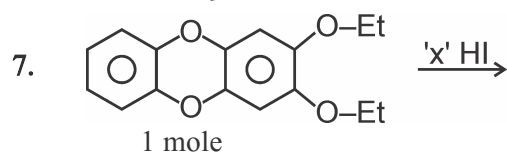
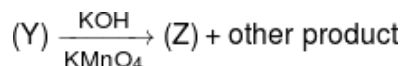
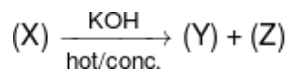
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1. If the number of following chemical species which form N_2O with dilute HNO_3 is x. Then predict the values of x.
Ag, Au, Fe^{+2} , Zn, S, $BaSO_3$, Hg, $CaCO_3$, I_2
 2. If the number of metal-metal bond, number of terminal CO ligand and number of bridging ligand in $Co_2(CO)_8$ (solid) are x, y & z respectively.
Find $(x + y + z)$.
 3. A fish swims in water at 300 K. In the process fish released 3000 Joule energy to water.
If $\Delta S_{fish} = -\frac{2}{3} \times 10^{-3} J/K$ for every joule of energy released by fish, find $\Delta S_{universe}$ in the process (in J/K)
 4. Determine degree of dissociation (in %) of 0.05 M NH_3 at $25^\circ C$ in a solution of pH = 11.
-

Space for Rough Work

5. The vapour pressures of two pure liquids A and B which form an ideal solution are 300 and 800 Torr, respectively, at temperature T. A liquid solution of A and B for which the mole fraction of A is 0.60 is contained in a cylinder closed by a piston on which the pressure can be varied. The solution is slowly vapourized at temperature T by decreasing the applied pressure. Find total vapour pressure (in torr) when last drop of liquid is remaining ?

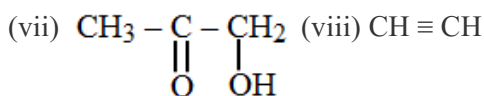
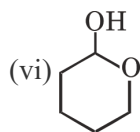
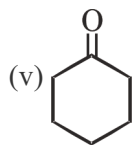
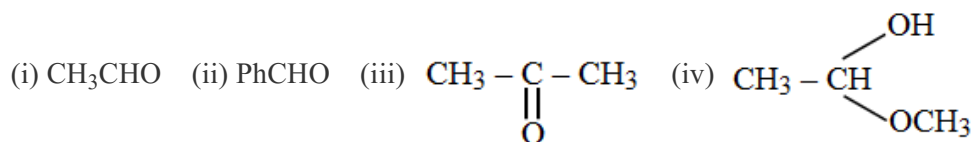
6. Find sum of the oxidation number of Iodine in (X) and (Z).

$\text{KI} + \text{conc. H}_2\text{SO}_4 \rightarrow (\text{X}) \text{ iodine containing product} + \text{other product}$



How many (x) moles of HI consumed in the given reaction ?

8. How many compounds will give positive Fehling test :



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Identify the correct statement(s) : (Where 'C' is constant of integration)

(A) $\int \frac{(x^5 - x)^{1/5}}{x^6} dx = \frac{5}{24} \left(1 - \frac{1}{x^4}\right)^{6/5} + C$

(B) $\int \frac{(x^5 - x)^{1/5}}{x^6} dx = -\frac{5}{24} \left(1 - \frac{1}{x^4}\right)^{6/5} + C$

(C) $\int (\tan^3 x + \tan^2 x + \tan x + 1) dx = \frac{(\tan x + 1)^2}{2} + C$

(D) $\int (\tan^3 x + \tan^2 x + \tan x + 1) dx = \frac{1}{2} \sec^2 x + \tan x + C$

Space for Rough Work

2. Given $I_n = \int_0^1 (1 - x^2)^n dx$ ($n = 0, 1, 2, \dots$), then which of the following is/are true ?
- (A) $\frac{I_{n+1}}{I_n}$ is independent of n (B) $\frac{I_{n+1}}{I_n} = \frac{2n+2}{2n+3}$
- (C) $I_k = \prod_{r=1}^k \left(\frac{2r}{2r+1} \right)$, ($k \in \mathbb{N}$) (D) there exist some n ($n \in \mathbb{N}$) for which $I_n \geq 1$
3. $\int \sqrt{1 + \operatorname{cosec} x} dx$ equals - (where C is integration constant)
- (A) $2\sin^{-1} \sqrt{\sin x} + C$ (B) $\sqrt{2}\cos^{-1} \sqrt{\cos x} + C$
- (C) $C - 2\sin^{-1} (1 - 2\sin x)$ (D) $\cos^{-1}(1 - 2\sin x) + C$
4. Let $A_n = \sum_{k=0}^{n-1} \frac{n}{k^2 + n^2 + (n+2)k + n+1}$ and $B_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$ for $n = 1, 2, 3, \dots$, then which of the following is/are true ?
- (A) $A_n < B_n$ (B) $A_n > B_n$ (C) $A_n < \frac{\pi}{3}$ (D) $B_n > \frac{\pi}{9}$
5. If $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 3}$, then which of the following statement(s) holds good ?
- (A) $f(x)$ is continuous at $x = 1$ (B) $f(x)$ is discontinuous at $x = 1$
- (C) $\lim_{x \rightarrow 2} f(x)$ exists finitely (D) $\lim_{x \rightarrow 2} f(x)$ does not exist
6. The function $f(x) = [\cos x] \cos x$ in $(0, 2\pi)$ -
- (A) is discontinuous at two points (B) is non differentiable at two points
- (C) has range $(0, 1]$ (D) is such that $f(f(x)) + 1 = 0$ has no real roots

Space for Rough Work

7. Let line L is normal to parabola $y^2 = 4x$ and line L intersect the parabola at A and B. If point P(α , β) is point of intersection of tangents at A and B then which of the following is/are correct ?
- (A) Locus of point P is symmetric w.r.t. x axis
 (B) Locus of point P is symmetric w.r.t. y axis
 (C) If $\alpha, \beta \in I$ then total number of possible ordered pair (α, β) is 2
 (D) If $\alpha, \beta \in I$ then total number of possible ordered pair (α, β) is 4
8. Let $f(x) = ax^9 + b\sin x + cx^2 \operatorname{sgn}(x) + \frac{e^x - e^{-x}}{e^x + e^{-x}}$ be defined on set of real number (where $a > 0, b, c \in \mathbb{R}$). If $f(-5) = 5, f(-2) = -3$, then the minimum number of zeroes of the equation $f(x) = 0$, is
- (A) 1 (B) 2 (C) 3 (D) 5
9. Given f and g are twice differentiable functions and $f' \neq 0, g' \neq 0$. If $f(x)g(x) = 1$, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ is equal to :
- (A) $\frac{-2g'(x)}{g(x)}$ (B) $\frac{g'(x)}{g(x)}$ (C) $\frac{2f'(x)}{f(x)}$ (D) $\frac{-f'(x)}{f(x)}$
10. If $\int \frac{7 + 6x^{13}}{\sqrt[3]{x^{17} - x^{30}}} dx = -\frac{p(1 - x^{13})^{\frac{q}{3}}}{2x^{\frac{r}{3}}} + C$ where $p, q, r \in \mathbb{N}$ and C is the constant of integration, then
- (A) $p = 3$ (B) $p = 7$ (C) $r - q = 12$ (D) $r - q = 8$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int_0^1 \frac{e^x dx}{\sqrt{1-x^2}} = A$ and $\int_0^\pi (e^{|\sin x|} + e^{|\cos x|}) dx$ is equal to λA , then the value of λ is -
2. The value of $\int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} dx$ is -
3. If $\int \frac{(3x^2 + 2x)dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} = \frac{1}{2} \tan^{-1}(g(x)) + C$, then the value of $(g(1) + g(0))$ is
4. Let $f(x)$ be a twice differentiable function such that $f''(0) = 2$, then the value of $\lim_{x \rightarrow 0} \frac{f(5x) - 3f(3x) + 2f(2x)}{x^2}$ is

Space for Rough Work

5. Let $g : \left[\frac{1}{2}, 1\right] \rightarrow (0, \infty)$, $g(x) = e^{-2x} f(x)$, where $g(x)$ is non constant decreasing and differentiable function and $g\left(\frac{1}{2}\right) = \frac{1}{e}$. Then the value of $\int_{\frac{1}{2}}^1 f(x) dx$ is
- [where $[x]$ denotes the greatest integer less than or equal to real number x]
6. The number of values of x , $x \in [-2, 3]$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous, is (where $[.]$ denotes greatest integer function) -
7. Let $\tan^{-1} x = \frac{1}{2} \tan^{-1} y = \frac{1}{3} \tan^{-1} z = \frac{\pi}{7}$, then the value of $\frac{1}{2}(x^2 + y^2 + z^2 - x^2 y^2 z^2)$, is
8. The maximum value of $f(x) = \frac{2x^2 + 7x + 3}{2x^2 + x + 1}$, is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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